ProxyPheno

An ML proxy-phenotype engine for novel trait discovery in dairy cattle

Technical report · Manju Vallayil · 24 October 2026
Data: Reynolds et al. (2021), Dryad doi:10.5061/dryad.kwh70rz1z · Code: github.com/manjuvallayil/proxypheno

Executive summary

Routine dairy measurements already contain latent signal about traits that are expensive or impossible to measure at population scale. ProxyPheno models the joint distribution of routine phenotypes, scores how unexpected each animal's multi-trait profile is, and groups animals by the trait that drives their unusualness. The proof of concept runs on the open deposit behind Reynolds et al. (2021), *Non-additive association analysis using proxy phenotypes identifies novel cattle syndromes* (Nature Genetics 53:949-954), whose co-authors include the LIC research director.

> **What works.** 80,027 animals, 19 traits, 22 interpretable signatures. Novelty rankings are stable (Spearman 0.988; top-5% Jaccard 0.783 across 10 bootstrap refits). The prospective calf study reproduces the PLCD4 bodyweight effect (-25.9%; 95% CI -62.3 to -39.4 kg). The chr14 casein region is recovered in milk and protein tails (q < 1e-100), including within breed.

> **What does not work as expected.** The deposited bodyweight yield deviations attenuate the 2021 bodyweight QTL effects, so discovery-population bodyweight enrichment is not a valid positive control. This is reported rather than hidden. The linear baseline is also strong: LightGBM improves mean R^2 only from 0.479 to 0.488.

1. Data

The deposit contains 19 growth, production and conformation traits for 80,027 animals, plus genotype matrices and supporting pedigree and GWAS files. Phenotypes are **pre-adjusted yield deviations**; herd, year and parity adjustment must not be re-run. The `rinverse` column is a reliability weight.

Block	Contents	Use here
`phen_1.*`	Bodyweight, stature, BCS	Discovery traits
`phen_4.*`	16 production and conformation traits	Discovery traits
`phen_3.1`	568 calves, 51 herds, four variants	Prospective positive control
`gen_3.*`	One SNP per prioritised mutation, 80,027 animals	Carrier enrichment
`other_1.*`	Dominance GWAS summary statistics	Mutation confirmation

The 12 prioritised mutations are the top dominance signals in the GWAS (e.g. chr2:107313998, B_D = 54.5; chr28:1312334, B_D = 55.2; chr29:44213160, B_D = 30.6). Genotype and phenotype tables are joined on `ID`; row order differs and is never assumed.

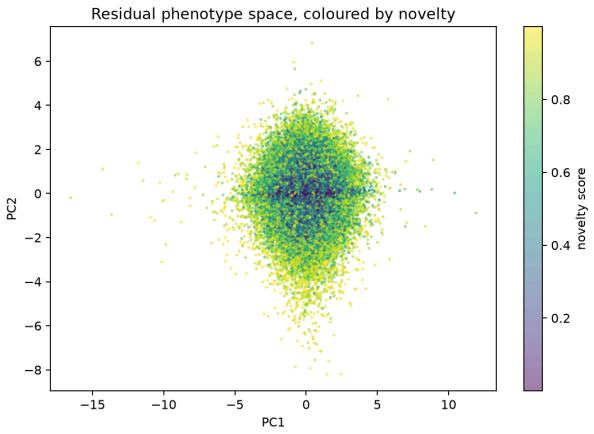
2. Method

1. Load the 19 traits; drop three with >10% missingness from the novelty model (`adaptability_milking`, `shed_temperament`, `farmer_opinion`).
2. Iteratively impute the remaining 16 traits and robust-scale them.
3. For each trait, fit a cross-fitted Ridge residual conditional on the other 15 traits. This is the proxy-phenotype signal: how unexpected a trait is given the rest of the profile.
4. Robust-scale the residuals; reduce with PCA (95% variance); score novelty with PCA-space Mahalanobis distance and IsolationForest, combined into a percentile score.
5. HDBSCAN found no density-separated clusters: the residual space is a single, approximately Gaussian continuum. Rather than force clusters, label each animal by its dominant residual trait and direction (`low_` / `high_`), with a |z| ≥ 3 threshold; the rest are `diffuse`.
6. Test the 12 prioritised mutations for carrier enrichment in trait tails, signatures and the top-5% novelty set, with Benjamini-Hochberg control within each analysis.
7. Compare Ridge, LightGBM and a small MLP on an identical 80/20 split.
8. Generate deterministic, template-based briefs from the structured results. The JSON contract is LLM-ready; no live model call is made and no number is generated.

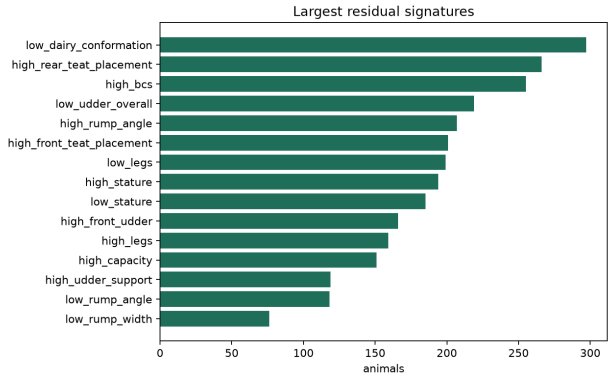
3. Results

3.1 Novelty and signatures

The 16-trait residual space is a continuum, not a set of natural clusters. The novelty score is therefore a ranking, and the signature is the interpretable unit. There are 22 signatures with at least 30 animals; 76,652 animals are diffuse. The largest signatures include `low_dairy_conformation` (n = 297), `high_rear_teat_placement` (n = 266) and `high_bcs` (n = 255).



Residual phenotype space, coloured by novelty score. The space is a single continuum.



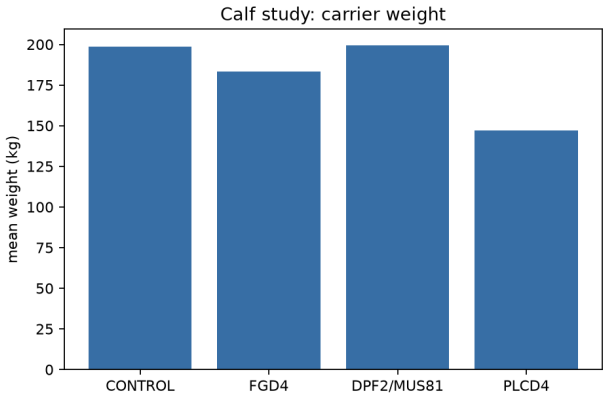
Largest dominant-trait signatures.

3.2 Stability

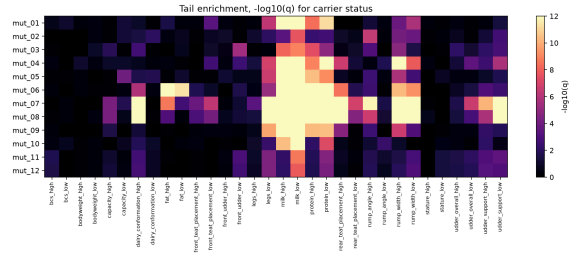
Ten bootstrap refits (80% subsamples) give a mean Spearman correlation of **0.988** (SD 0.001) and a top-5% Jaccard overlap of **0.783**. The ranking is not an artefact of one fit.

3.3 Positive controls

The prospective calf study is the clean test: PLCD4 carriers weigh **147.1 kg** versus **198.4 kg** for controls, a **-25.9%** difference (95% CI -62.3 to -39.4 kg; 9 carriers vs 526 controls). This reproduces the published effect. In the discovery population, the chr14 casein region (mut_06/07/08) is strongly enriched in milk and protein tails (e.g. mut_07 high milk OR 6.8, q ≈ 1e-170) and persists within Holstein-Friesian and Jersey.



Prospective calf study: PLCD4 carriers weigh 25.9% less than controls.

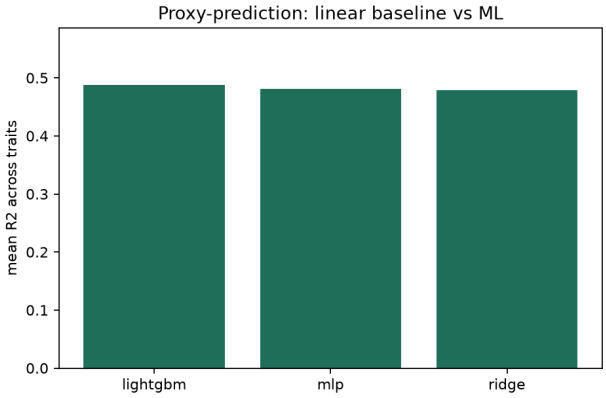


Carrier enrichment in trait tails, -log10(q). The chr14 casein block dominates milk and protein.

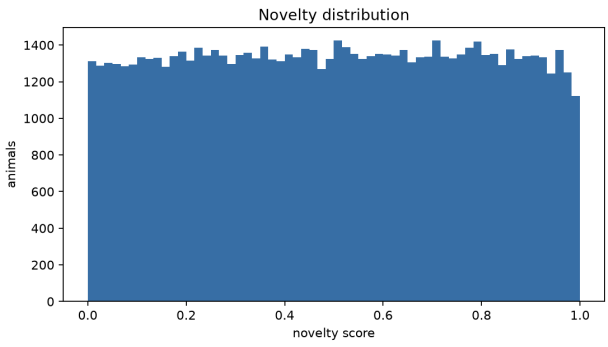
> **Bodyweight caveat.** The deposited `yd_trait` is attenuated for the 2021 bodyweight loci (PLCD4 homozygotes differ by roughly 3 kg, not 50 kg). Discovery-population bodyweight enrichment is therefore not a valid positive control, and is not claimed.

3.4 Baseline vs ML

On an identical 80/20 hold-out, LightGBM reaches a mean R^2 of **0.488**, the MLP **0.481** and Ridge **0.479**. LightGBM wins on 14 of 16 traits, but the linear baseline is already strong; the ML gain is modest and consistent, not a step change. That is the honest result.



Mean R^2 across 16 traits on the same split.



Novelty score distribution; the top 5% threshold is 0.946.

4. What this is not

- Not a new causal variant. The output is a ranked set of discovery-support hypotheses.

- Not a claim that ML beats established genetic models. The gain here is small.
- Not a diagnosis. GenAI-style briefs translate structured results; they do not diagnose.
- Not a claim that the open deposit equals LIC production data.
- Not production-ready. Held-out herd/year validation and conformal coverage are next.

5. Path to LIC data

The design assumes 10^5-10^6 animals. The pipeline is phenotype-first, uses columnar storage, and reads genotypes in chunks; it does not require the full genotype matrix in memory. The next steps on internal data are: (1) reproduce the conditional-residual baseline on LIC phenotypes, (2) agree trait priorities with genetics and animal health, (3) add held-out herd/year validation and conformal intervals, (4) nominate one candidate phenotype for formal validation, and (5) connect the brief layer to an internal LLM with the same structured contract.

6. Reproducibility

The public repository contains the package, tests, precomputed artifacts and figures. Commands: `make setup`, `make test`, `make pipeline`, `make compare`, `make app`. The Dryad deposit is not redistributed; download it manually and place it at `data/raw/` or a sibling `dryad/` folder. The pipeline runs in about 21 seconds after download; the model comparison adds about one minute.

Anchor: Reynolds, E.G.M., Neeley, C., Lopdell, T.J., Keehan, M., Dittmer, K., Harland, C.S., Couldrey, C., Johnson, T.J.J., Tiplady, K., Worth, G., Walker, M., Davis, S.R., Sherlock, R.G., Carnie, K., Harris, B.L., Charlier, C., Georges, M., Spelman, R.J., Garrick, D.J., Littlejohn, M.D. (2021). Non-additive association analysis using proxy phenotypes identifies novel cattle syndromes. Nature Genetics 53, 949-954. doi:10.1038/s41588-021-00872-5.